INCH-POUND

A-A-54247A
9 September 1994
SUPERSEDING
A-A-54247
21 September 1990

COMMERCIAL ITEM DESCRIPTION

REMOVER, DENTAL CROWN (Morrell)

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers a crown remover suitable for use during dental operative procedures.

Salient characteristics:

Material: shaft and points. The shaft shall be fabricated from type 303, 416, or 440A stainless steel. The points shall be fabricated from type 440A stainless steel and shall be in accordance with American Society for Testing and Materials (ASTM) Designation F899 for composition only when tested as specified herein.

Material: weight, sleeve and cap. The weight and sleeve shall be fabricated of Class 3, austenitic stainless steel.

Hardness. After proper stress relieving, the points shall have a Rockwell hardness of C50 and C55 when determined in accordance with American Society for Testing and Materials (ASTM) Designation E18 (Rockwell) or E92 (Vickers).

Design and dimensions. The crown remover shall be of the type known as "Morrell" and shall be suitable for use during dental operative procedures. The crown remover shall be furnished with one contra-angle point (Type 1 or A) and one right-angle point (Type 2 or B). The crown remover shall be in accordance with Figures 1 or 2 for design and dimensions.

The crown remover shall be packaged in a close fitting container and packaged to prevent damage to the components.

AMSC N/A FSC 6520

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Construction. The crown remover shall consist of a cap, shaft, weight, and two interchangeable points. The cap shall be knurled, and shall be securely and permanently attached to the proximal end of the shaft except the alternate cap shall be removable by unthreading. The weight shall move freely along the shaft between the cap and the pinched area of the point (stops). The weight shall weigh 75 ± 10 grams. The cap shall withstand the impact test without failure, when tested as specified herein.

Finish. The points shall have a satin or bright finish, except for the sharpened working end which may have a fine honed finish. The remainder of the crown remover shall have a satin or bright finish, except for the knurled portion.

Corrosion resistance. The shaft and points shall meet the copper sulfate test requirements of American Society for Testing and Materials (ASTM) Designation F1089. The entire assembled crown remover shall meet the boil test requirements of American Society for Testing and Materials (ASTM) Designation F1089.

Identification marking. The shaft and points shall be permanently and legibly marked. Contra-angle points (Type 1 or A) shall be marked with the number "1" or the letter "A" and straight angle points (Type 2 or B) shall be marked with the number "2" or the letter "B". Each crown remover handle shall be marked with the name or registered trademark of the manufacturer and contractor, if other than manufacturer. Each crown remover handle shall be marked with the word "stainless". As an alternate to the word "stainless", the use of a registered trade name identifying the steel as stainless is acceptable. Instruments of foreign manufacture shall be permanently and legibly marked with the country of origin, together with the name or registered trademark of the contractor and manufacturer.

Workmanship. The crown remover shall be free from burrs, nicks, rough edges and other defects which may detract from its appearance or impair its serviceability, proper function, or intended use.

Unit. Each (EA). One crown remover, as specified, constitutes one unit. Each crown remover shall be supplied in a suitable commercial package.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after the delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Examination. Examination shall be conducted to determine compliance with specification requirements.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as specified. Unit of product for examination purposes shall be one crown remover. Sampling shall be inspection level S-2. Acceptable Quality Level (AQL) shall be zero for all sample sizes.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified. The unit of product for test purposes shall be one crown remover. The Inspection Levels and Acceptable Quality Levels (AQLs) shall be as indicated below.

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Test criteria.

Characteristic	Inspection level	AQL
Chemical analysis	1/	2/
Hardness	S-2	2/
Dimensions	S-2	2.5
Impact test	S-2	2/
Corrosion resistance	S-2	2/

1/ A sufficient quantity of each lot of raw material used in the fabrication of the crown removers shall be selected for testing.

2/ The acceptance number shall be zero for all sample sizes.

Chemical analysis. In the absence of certificates of quality, chemical analysis of the individual components of the crown remover shall be determined in accordance with method 111 of FED-STD-151.

Impact test. The number 2 (straight angle) point shall be securely threaded in the socket of the crown remover. The crown remover shall be securely fixed at a point around the threaded knurled sleeve, or above, in such a manner that the fixation will not interfere with the sliding weight as it moves vertically upward on the shaft, from cap to pinched area of the point. A 20 ± 1 pound weight shall be secured to the sliding weight.

The sliding weight, with its added weight, shall be pushed up the shaft until the sliding weight makes contact with the bottom of pinched area of the point, and then it shall be released. This impact test shall be performed five times, and then the cap shall be examined for evidence of movement of loosening of cap.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

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NOTE: The following National Stock Number and its respective Item Identification is covered by this document:

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6520-00-551-6150	REMOVER, DENTAL CROWN (Morrell)

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project No. 6520-2580
Location: preston/crown/dawne

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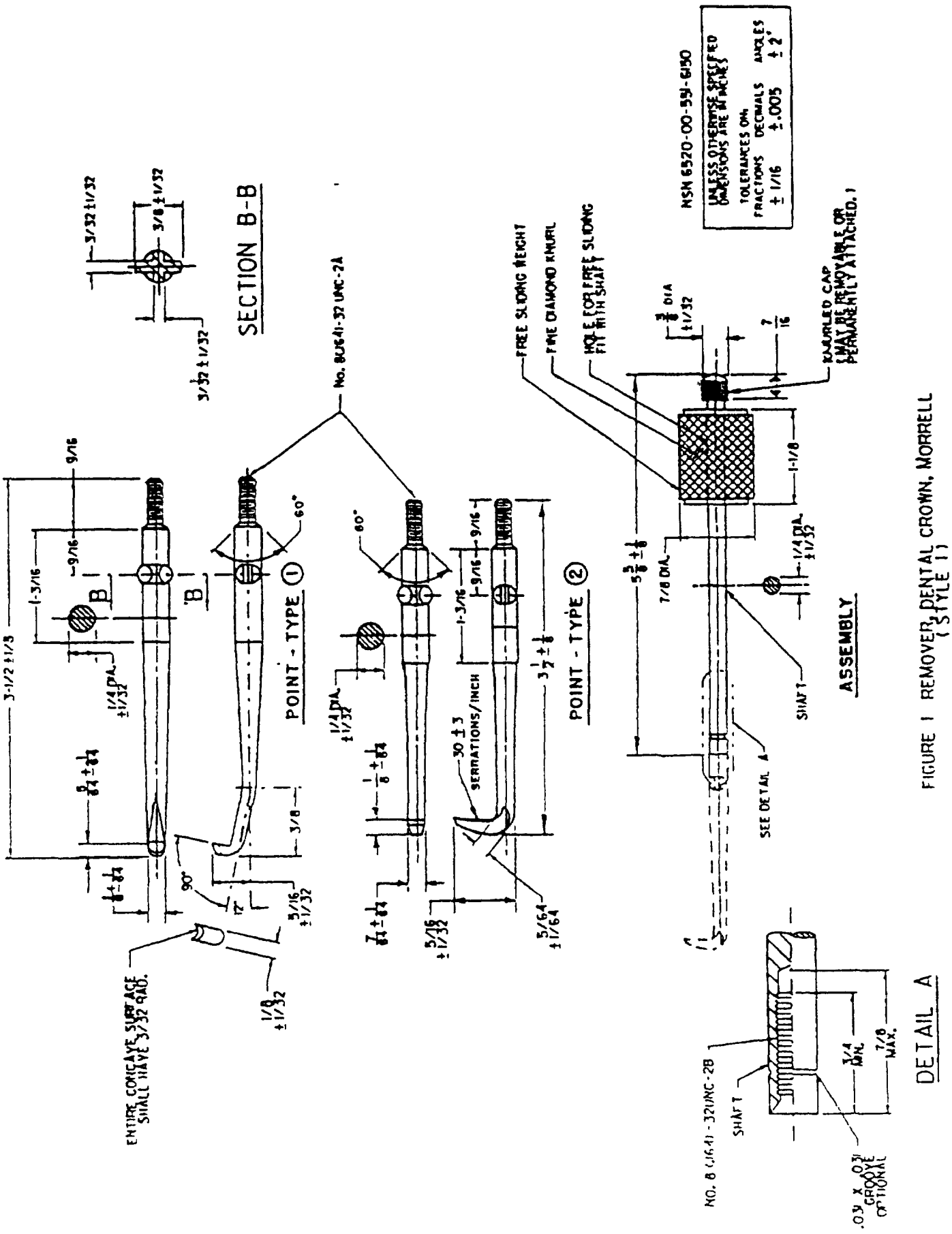


FIGURE 1 REMOVER, DENTAL CROWN, MORRELL (STYLE 1)

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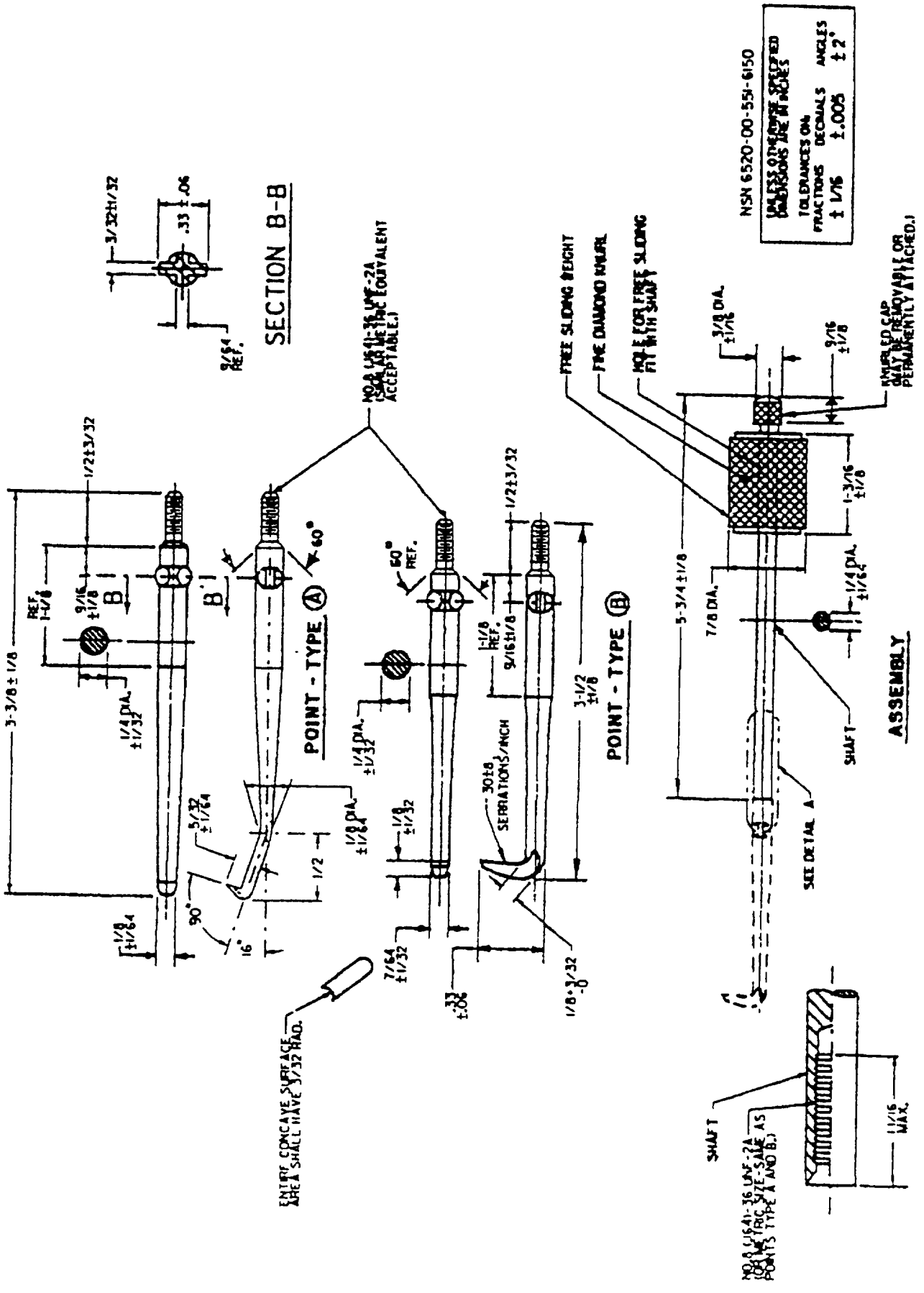


FIGURE 2 REMOVER, DENTAL CROWN, MORRELL (STYLE 2)

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STANDARDIZATION DOCUMENT [illegible]

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER: A-A-54247A	2. DOCUMENT DATE (YYMMDD): 94/09/09

3. DOCUMENT TITLE: REMOVER, DENTAL CROWN (Morrell)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Defense Personnel Support Center ATTN: DPSC-MAP	(1) Commercial: (215) 737-8431	(2) AUTOVON: 444-8431
ADDRESS (Include Zip Code) 2800 South 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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